HE, AHE, CLAHE, NICLAHE: A Journey Towards Intelligent Image Enhancement

(based on https://doi.org/10.18240/ijo.2025.12.02 "An improved neighbourhood-based contrast limited adaptive histogram equalization method for contrast enhancement on retinal images")

1.Histogram Equalization (HE)

What it does: Takes the whole image and stretches its brightness histogram to fill the full 0–255 range.

Problem: Global approach – treats dark and bright regions the same. Often over-amplifies noise in uniform areas or washes out already-bright regions. Not suitable for images with mixed local characteristics (like medical images).

2.Adaptive Histogram Equalization (AHE)

What it does:

- Divides the image into small tiles and performs histogram equalization on **each tile independently**.
- Each tile gets its own stretching based only on the pixels inside that tile.

Improvement over HE: Local adaptation – different regions get different enhancement based on their own content.

Problem:

- **Noise amplification** – in nearly uniform tiles (e.g., dark background), even tiny random brightness variations get stretched dramatically, creating unnatural speckles.
- **Blocky artefacts** – since each tile is processed independently with no blending, tile boundaries become clearly visible in the final image, giving it a mosaic or checkerboard appearance.

3.CLAHE (Contrast Limited Adaptive Histogram Equalization)

What it does:

- Divides the image into small tiles and equalizes each tile separately (like AHE).
- **Clips the histogram** in each tile to limit noise amplification – no single brightness can dominate.
- **Smoothly blends** tile edges using bilinear interpolation to eliminate blocky artefacts.

Improvement over AHE:

- The clip limit acts as a "brake" on enhancement, preventing noise from being over-amplified in flat regions.
- Blending removes the visible tile boundaries.

Limitation: The **clip limit is fixed** for all tiles. A dark, flat background gets the same amount of "braking" as a detail-rich vessel area. This is not optimal – flat areas need more enhancement; textured areas need less.

4.NICLAHE (Neighbourhood-based Improved CLAHE)

What it does:

Everything CLAHE does, **plus**:

- For each tile, looks at its **neighboring tiles** (3×3 block) to estimate the local contrast level.
- **Adapts the clip limit per tile** using this neighborhood information:
    - **Flat regions** (low neighborhood contrast) → **higher clip limit** → stronger enhancement to bring out hidden details.
    - **Textured regions** (high neighborhood contrast) → **lower clip limit** → milder enhancement to avoid over-amplifying noise.

Key difference from CLAHE: The clip limit is **dynamic and content-aware**, not fixed. It uses a smooth mathematical function (sigmoid) to adjust the limit based on the average local standard deviation of the surrounding tiles.

Summary Table

Method	Local?	Clip Limit?	Blending?	Adapts to Content?	Main Problem
HE	No (global)	No	No	No	Washes out or darkens regions unevenly
AHE	Yes (tiles)	No	No	Partially (per tile)	Amplifies noise + blocky artefacts
CLAHE	Yes (tiles)	Yes, fixed	Yes	Partially (per tile)	Same clip limit for all regions
NICLAHE	Yes (tiles)	Yes, adaptive	Yes	Yes – uses neighbourhood contrast	More complex to implement

The Evolution Summary:

- **HE:** Stretch the whole image's contrast globally.
- **AHE:** Stretch each small region independently.
- **CLAHE:** Stretch each region, but put a brake on noise and blend edges smoothly.
- **NICLAHE:** Smart brake – adjust the braking force based on what the neighboring regions look like.

5. Why these matters for retinal images

Retinal photographs often have:

- Large dark backgrounds,
- Thin, low-contrast blood vessels,
- Bright optic disc regions.

A fixed clip limit would either fail to reveal the faintest vessels or introduce noise in the background. NICLAHE automatically adjusts: the background gets a strong boost to pull out hidden vessel details, while the optic disc area is treated more conservatively to preserve its natural appearance. This makes the enhanced image much more useful for diagnosing diseases like diabetic retinopathy.